Hands-On-Activity No. 4.2

Hands-on Activity 4.2: Arrays

Course Code: CPE 007	**Program: Computer Engineering**
Course Title: Programming logic and Design	**Date Performed: August 18, 2026**
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1. Objective(s)

This activity aims to develop programming skills using arrays.

2. Intended Learning Outcomes

2.1 Understand array data structure

2.2 Declare array, initialize an array and refer to individual elements of an array

3. Materials and Equipment

Internet connection
Personal Computer

4. Output

1. Given the size of an array which is 10, and the elements such as 19, 3, 15, 7, 11, 9, 13, 5, 17 and 1, create a program that will display the following output:

```
Element    Value    Histogram
      0       19    ********************
      1        3    ***
      2       15    ***************
      3        7    *******
      4       11    ***********
      5        9    *********
      6       13    *************
      7        5    *****
      8       17    *****************
      9        1    *
```

Answer:

```
#include <iostream>
using namespace std;

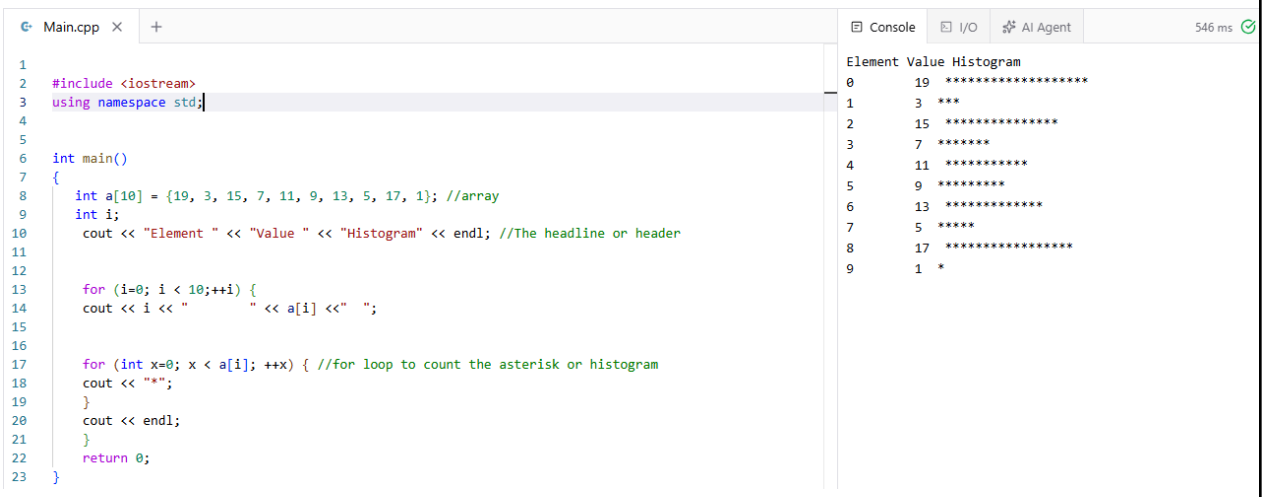
int main()
{
  int a[10] = {19, 3, 15, 7, 11, 9, 13, 5, 17, 1}; //array
  int i;
```

```
   cout << "Element " << "Value " << "Histogram" << endl; //The headline or header


   for (i=0; i < 10;++i) {
   cout << i << "        " << a[i] <<"  ";


   for (int x=0; x < a[i]; ++x) { //for loop to count the asterisk or histogram
   cout << "*";
   }
   cout << endl;
   }
   return 0;
}
```

Screen Shot:



2. Given the following data, create a program that summarizes the number of each type. Use array responses for the 40 element array of student's responses. Such as

int responses[RESPONSE_SIZE] = { 1, 2, 6, 4, 8, 5, 9, 7, 8, 10, 1, 6, 3, 8, 6, 10, 3, 8, 2, 7, 6, 5, 7, 6, 8, 6, 7, 5, 6, 6, 5, 6, 7, 5, 6, 4, 8, 6, 8, 10}

Answer:

```
#include <iostream>
using namespace std;

int main()
{
  //the 40 responses of the students
   int responses[40] = { 1, 2, 6, 4, 8, 5, 9, 7, 8, 10,
                         1, 6, 3, 8, 6, 10, 3, 8, 2, 7,
                         6, 5, 7, 6, 8, 6, 7, 5, 6, 6,
                         5, 6, 7, 5, 6, 4, 8, 6, 8, 10 };

   int counts[11] = {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}; // to hold the countings 1-10
```

```
  //will count how many times each number appears
  for (int i = 0; i < 40; ++i) {
     int currentRating = responses[i];
     counts[currentRating] = counts[currentRating] + 1;
  }


  cout << "Rating" <<"  "<< "Count" <<"   "<< "Histogram" << endl;

 //for loop for rating
  for (int rating = 1; rating <= 10; ++rating) {

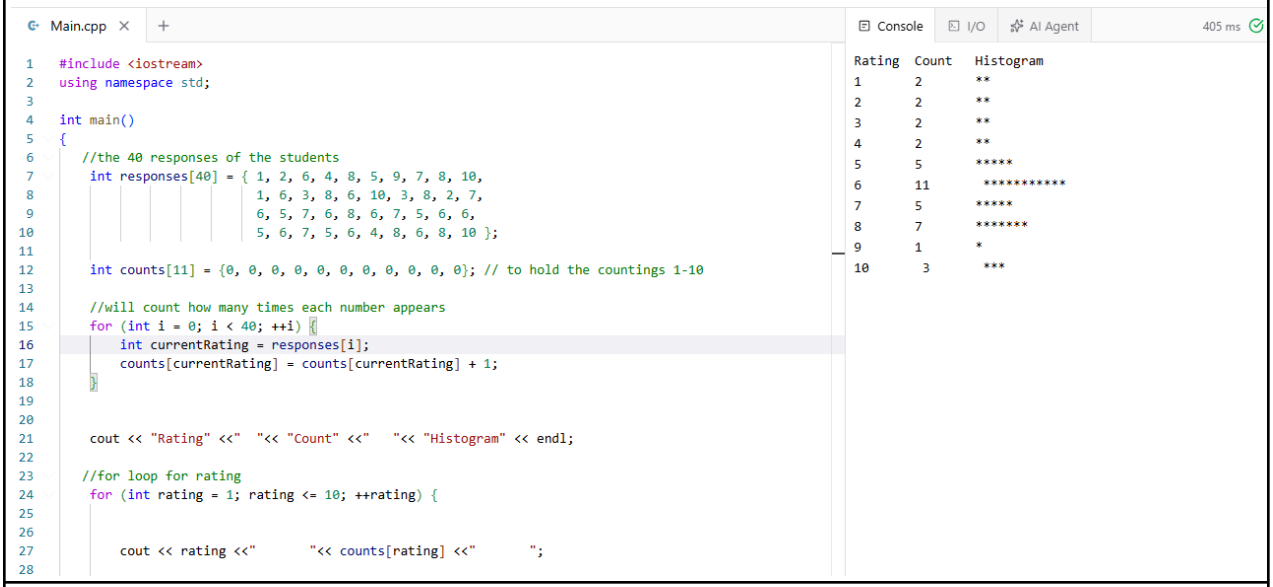

     cout << rating <<"       "<< counts[rating] <<"       ";

    //for loop for count
     for (int x = 0; x < counts[rating]; ++x) {
        cout <<"*";
     }

     cout << endl; //ends line to start the next
  }

  return 0;
}
```

Screenshot



5. Conclusion

Learning about array

6. Rubric